Homework 1

● Graded

Student

Toby Hansen

Total Points

81 / 92 pts

Question 1

ESP **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts totally wrong or missing solution

(a) 3 points

– 3 pts No solution or totally wrong

– 0.5 pts Correct proof but missing minor details, e.g., explicit statement of commutative rule

– 1 pt Correct proof but missing a major or some minor steps

– 1.5 pts Errors in proof but some correct ideas

– 2 pts Too many errors in proof but a few correct ideas

(b) 3 points

– 3 pts No solution or totally wrong

– 0.5 pts Correct proof but missing minor details, e.g., explicit statement of commutative rule

– 1 pt Correct proof but missing a major or some minor steps

– 1.5 pts Errors in proof but some correct ideas

– 2 pts Too many errors in proof but a few correct ideas

(c) 4 points

– 4 pts No solution or totally wrong

– 1 pt Correct proof but missing minor details

– 1.5 pts Correct proof but missing a major or some minor steps

– 2 pts Errors in proof but some correct ideas

– 3 pts Too many errors in proof but a few correct ideas

Question 2

DSP

10 / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts totally wrong or missing solution

(a) 4 points

– 4 pts No solution or totally wrong

– 1 pt Not in proper required format

– 1 pt minor errors (see solution key for one correct answer)

– 3 pts major errors (see solution key for one correct answer)

(b) 3 points

– 3 pts No solution or totally wrong

– 1 pt Not in proper required format

– 1 pt minor errors (see solution key for one correct answer)

– 2 pts major errors (see solution key for one correct answer)

– 1 pt Missing some columns

(c) 3 points

– 3 pts No solution or totally wrong

– 2 pts very wrong

– 1 pt Correct counterexample but did not provide evaluation of proposition with those truth settings as required

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– 0 pts Click here to replace this description.

Question 3

P1.1.8

8 / 10 pts

– **0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– **10 pts** totally wrong or missing solution

(a) 3 points

– **3 pts** missing solution

– **2 pts** Correct values with no justification.

– **2 pts** Argument with incorrect solution

– **1 pt** Gave incorrect value of n

✔ – **1 pt** Argument that $\{1\}$ and $\{2\}$ are only ones is not fully correct

3 But why must $n-1$ and 1 be distinct? If they're not then your argument fails (and it should fail for $n=2$, as your solution claims).

– **0.5 pts** Missing set $\{1\}$

– **0.5 pts** Missing set $\{2\}$

– **0.5 pts** Incorrect set notation

– **0.5 pts** missing $\{n\}$ or $\{1, n-1\}$

(b) 4 points

– **4 pts** missing solution

– **3 pts** only half the sets included are correct

– **1 pt** missing two sets

– **0.5 pts** missing one set

– **1 pt** included two incorrect sets

– **0.5 pts** included one incorrect set

(c) 3 points

– **3 pts** missing solution

– **2 pts** argument has serious correctness issues

✔ – **1 pt** argument is mildly incorrect

1 This is not what we asked you to answer - the question mentions nothing about S and T being the unique set for their sum.

It would have been correct to say "if $S \neq T$ and $sum(S) = sum(T)$", since this is what the question asked, and is also what the rest of your proof correctly shows.

2 Please assign pages next time.

Question 4

Modified P1.1.10

10 / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts totally wrong or missing solution

(a) 2.5 points

– 2.5 pts Missing or totally wrong solution

– 2 pts Got (i) wrong but gave some justification

– 1.5 pts Got (i) correct, but gave no or totally incorrect justification

– 2 pts Got (i) correct and gave only partial justification in (ii) [Correct justification needed to give an example with the number and ALSO, prove the number is minimal]

– 0.5 pts Correct but weak justification for the empty set case.

(b) 2.5 points

– 2.5 pts Missing or totally wrong solution

– 2 pts Got (i) wrong but gave some justification

– 1.5 pts Got (i) correct, but gave no or totally incorrect justification

– 2 pts Got (i) correct and gave only partial justification in (ii) [Correct justification needed to give an example with the number and ALSO, prove the number is minimal]

– 0.5 pts Weak justification for the empty set case.

(c) 2.5 points

– 2.5 pts Missing or totally wrong solution

– 2 pts Got (i) wrong but gave some justification

– 1.5 pts Got (i) correct, but gave no or totally incorrect justification

– 2 pts Got (i) correct and gave only partial justification in (ii) [Correct justification needed to give an example with the number and ALSO, prove the number is minimal]

– 0.5 pts Weak justification for the empty set case.

(d) 2.5 points

– 2.5 pts Missing or totally wrong solution

– 2 pts Got (i) wrong but gave some justification

– 1.5 pts Got (i) correct, but gave no or totally incorrect justification

– 2 pts Got (i) correct and gave only partial justification in (ii) [Correct justification needed to give an example with the number and ALSO, prove the number is minimal]

– 0.5 pts Weak justification for the empty set case.

Question 5

P1.4.3 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts totally wrong or missing solution

(a) 4 points

– 4 pts missing answer

– 3 pts incorrect answer, i.e., totally or very incorrect justification

– 2 pts some partially valid justification

– 0.5 pts incorrect answer with totally valid justification (typo in the answer)

(b) 3 points

– 3 pts missing answer

– 2 pts incorrect answer with no justification

– 1 pt incorrect answer with some partially valid justification

– 0.5 pts incorrect answer with totally valid justification(typo in the answer)

(c) 3 points

– 3 pts missing answer

– 2 pts incorrect answer with no justification

– 1 pt incorrect answer with some partially valid justification

– 0.5 pts incorrect answer with totally valid justification(typo in the answer)

Please assign pages when submitting homework. In the future, points will be deducted for unassigned pages.

Question 6

P1.5.10 **7.5** / 10 pts

– **0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– **10 pts** totally wrong or missing solution

(a) 3 points

– **3 pts** missing solution

– **3 pts** incorrect solution with missing/incorrect explanation

– **2 pts** incorrect solution but with reasonable explanation

– **2 pts** correct solution but missing/bad explanation

– **1 pt** almost correct solution with almost correct explanation

– **0.5 pts** everything correct except for missing the empty set

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(b) 3 points

– **3 pts** missing solution

– **3 pts** incorrect solution with missing/incorrect explanation

– **2 pts** incorrect solution but with reasonable explanation

– **2 pts** correct solution but missing/bad explanation

– **1 pt** almost correct solution with almost correct explanation

– **0.5 pts** everything correct except for missing the empty set

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(c) 4 points

– **4 pts** missing solution

– **4 pts** incorrect solution with missing/incorrect explanation

✔ – **2.5 pts** incorrect solution but with reasonable explanation

– **2.5 pts** correct solution but missing/bad explanation

– **1 pt** almost correct solution with almost correct explanation

– **0.5 pts** everything correct except for missing the empty set

Question 7

P1.7.6 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts totally wrong or missing solution

(a) 8 points

– 8 pts missing soluton

– 6 pts incorrect counterexample without explanation

– 4 pts correct counterexample but doesn't work through any details as to why it's a counterexample

– 2 pts correct counterexample but doesn't work through enough details as to why it's a counterexample

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(b) 2 points

– 2 pts missing solution

– 1 pt unclear or incorrect explanation

Question 8

P2.1.4 **10** / 10 pts

✔ **– 0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– 10 pts totally wrong or missing solution

(a) 6 points

– 6 pts missing solution

– 3 pts missing all items in $A \times (B \times C)$

– 2 pts list of items in $A \times (B \times C)$ very incorrect

– 1 pt list of items in $A \times (B \times C)$ slightly incorrect

– 3 pts missing all items in $(A \times B) \times C$

– 2 pts list of items in $(A \times B) \times C$ very incorrect

– 1 pt list of items in $(A \times B) \times C$ slightly incorrect

– 4 pts only listed items in $A \times B \times C$ instead

(b) 4 points

– 4 pts missing solution

– 2 pts unclear or incorrect explanation

– 1 pt small error in explanation due to mistake in part (a)

– 1 pt insufficient/unclear explanation

Question 9

EC: P1.6.9 **3.5** / 10 pts

– **0 pts** Correct

Caveat for incorrect answers: Our marking will often stop after finding some errors and deducting points for them. This permits awarding partial credit for having some correct content. This means that the marking might not identify ALL of the errors in your solution.

– **10 pts** totally wrong or missing solution

(a)

– **5 pts** (a) solution totally missing or incorrect

– **2.5 pts** (a) (i) solution totally missing or incorrect

✔ – **2.5 pts** (a) (ii) solution totally missing or incorrect

– **1.5 pts** (a) (i) truth table had some errors

– **1.5 pts** (a) (ii) deductive proof had some errors

– **0.5 pts** (a) (ii) deductive proof correct but did not mention at the end that it was necessary to check that the setting found actually was true for both the initial premises.

(b)

– **5 pts** (b) solution totally missing or incorrect

– **0.5 pts** (b) (i) solution totally missing or incorrect

– **0.5 pts** (b) (ii) solution totally missing or incorrect

✔ – **3 pts** (b) (iii) missing or totally incorrect

– **0.5 pts** b (iii) deductive proof was “correct” but missing some steps

– **1.5 pts** b (iii) deductive proof was incorrect but had some correct ideas

– **1 pt** b(iii) (i) and.or (ii) were incorrect but deductive proof for the written (i) and (ii) was correct

✔ – **1 pt** b (iv) totally missing or incorrect.

Question 10

EC: Latex **2** / 2 pts

✔ – **0 pts** latex

– **2 pts** not latex

No questions assigned to the following page.

250 Homework #1 (Fall 2026)

Toby Hansen *

This assignment is due at 11:59 p.m. on Friday 25 September 2026, to be submitted on Gradescope. Late assignments (with penalty, unless excused) will be accepted until 11:59 p.m. on Saturday 26 September 2026, and not after. Please see the syllabus for more details on homework grading and late penalties.

When submitting to Gradescope, please do not forget to assign pages to problems.

All problems, with the exception of the first two, originally come from the textbook. Some of them have been modified. In case of doubt, the version printed here is the one you should follow.

The main Canvas Homework page gives detailed submission guidelines, split into 6 major areas, listed below. Please see the Canvas Homework page for the complete guideline details.

1. **Proper page formatting.** (Use of Latex is recommended).
2. **Proper proof writing and formatting.**
   Proofs are marked on understandability, as well as correctness.
3. **Only use notation taught in class.**
4. **Use specified proof structures.**
5. **Properly declare collaborations and resources used.**
   Fill out the brief individual form for each question.
   This will be available as a survey on Canvas.
6. **Proper Use of LLMs.**

Solutions that do not follow guidelines 1-4 will not be marked. For the first two assignments, such submissions will be returned for rewriting and resubmission. In later assignments points will be deducted for solutions not following guidelines 1 and 2; solutions not following 3 and 4 will not be marked. Questions that do not have their seperate acknowledgment form (guidelines 5 and 6) completed on gradescope will not be marked.

See the Canvas Homework web page for detailed instructions on how to format solutions. Solutions that do not follow those formatting instructions will be returned for rewriting and resubmission. In later assignments there will be points deducted for such formatting errors.

*Collaborated with Nobody.

No questions assigned to the following page.

Problem 1 [10 pts]

Prove the correctness of (a), (b) and (c) using equational sequence proofs and not using a truth table. See the comments after (c) for instructions on how to format the proofs.

(a) Prove $(x \leftrightarrow y) \leftrightarrow (\neg x \leftrightarrow \neg y)$

Note/Hint: This is a warm up problem. Start with the definition of $\leftrightarrow$ given in E1(b) and then use E3(a), the contrapositive rule, twice, then the commutative rule, then E1(b) again.

(b) Prove $(p \wedge (p \oplus q)) \leftrightarrow (p \wedge \neg q)$ using an equational sequence proof.

(c) Prove $\Big((p \rightarrow r) \wedge (q \rightarrow r)\Big) \leftrightarrow ((p \vee q) \rightarrow r)$ using an equational sequence proof.

For each part, the proof structure should be the same as in part 2 of the *Equational and Deductive Sequence Proofs Study Guide.* That is, each line should provide a justification as to how it was derived from the previous one(s).

The justifications should only be rules listed in the study guide.

Proofs not following this structure will be ignored and marked incorrect.

A hint for part (c) is to start by using the definition of $\rightarrow$ given in E1 in the study guide.

Further Restrictions on how the solution must be formatted.

- Each line may reference only ONE justification rule (although that rule may be applied multiple independent times to different parts of one line).
- EVERYTHING must be proven. For example, this would not be a correct proof line:

$$(a \vee b) \quad \leftrightarrow \quad (\neg\neg b \vee \neg\neg a) \qquad \text{E9. Double negation, applied twice}$$

A correct proof would need to acknowledge the use of commutativity and justify it on separate line, e.g., as

$$\begin{array}{lll} (a \vee b) & \leftrightarrow & (b \vee a) \qquad \text{E3(b) Commutative Rule} \\ & \leftrightarrow & (\neg\neg b \vee \neg\neg a) \qquad \text{E9. Double negation, applied twice} \end{array}$$

No questions assigned to the following page.

Solution:

For (a) and (b) we have started you off with the first line and the start of the last line as well as the structure of the intermediate lines (with accompanying latex formatting in the provided template).

(a)

$(x \leftrightarrow y)$	$\leftrightarrow$	$((x \rightarrow y) \wedge (y \rightarrow x))$	E1(b). Def of $\leftrightarrow$
	$\leftrightarrow$	$((y \rightarrow x) \wedge (x \rightarrow y))$	E3(a). Commutative Rule
	$\leftrightarrow$	$((\neg x \rightarrow \neg y) \wedge (x \rightarrow y))$	E11. Contrapositive Rule
	$\leftrightarrow$	$((\neg x \rightarrow \neg y) \wedge (\neg y \rightarrow \neg x))$	E11. Contrapositive Rule
	$\leftrightarrow$	$((\neg y \rightarrow \neg x) \wedge (\neg x \rightarrow \neg y))$	E3(a). Commutative Rule
	$\leftrightarrow$	$(\neg y \leftrightarrow \neg x)$	E1(b). Def of $\leftrightarrow$
	$\leftrightarrow$	$(\neg x \leftrightarrow \neg y)$	E3(d). Commutative Rule

(b)

$(p \wedge (p \oplus q))$	$\leftrightarrow$	$p \wedge ((p \wedge \neg q) \vee (\neg p \wedge q))$	E1(d). Definition of $\oplus$
	$\leftrightarrow$	$(p \wedge (p \wedge \neg q)) \vee (p \wedge (\neg p \wedge q))$	E7(a). Distributive Law
	$\leftrightarrow$	$(p \wedge (p \wedge \neg q)) \vee ((p \wedge \neg p) \wedge q)$	E2(a). Associative Rule
	$\leftrightarrow$	$(p \wedge (p \wedge \neg q)) \vee (0 \wedge q)$	ESP2. $(p \wedge \neg p) \leftrightarrow 0$
	$\leftrightarrow$	$(p \wedge (p \wedge \neg q)) \vee 0$	E4(a). Zero Rule
	$\leftrightarrow$	$p \wedge (p \wedge \neg q)$	E4(b). Zero Rule
	$\leftrightarrow$	$(p \wedge p) \wedge \neg q$	E2(a). Associative Rule
	$\leftrightarrow$	$p \wedge \neg q$	E6(a). Equal Operands

(c)

$((p \rightarrow r) \wedge (q \rightarrow r))$	$\leftrightarrow$	$((\neg p \vee r) \wedge (\neg q \vee r))$	E1(a). Definition of $\rightarrow$
	$\leftrightarrow$	$((r \vee \neg p) \wedge (r \vee \neg q))$	E3(b). Commutative Rule
	$\leftrightarrow$	$r \vee (\neg p \wedge \neg q)$	E7(b). Distributive Law
	$\leftrightarrow$	$r \vee \neg(\neg\neg p \vee \neg\neg q)$	E10(a). DeMorgan's Law
	$\leftrightarrow$	$r \vee \neg(p \vee q)$	E9. Double Negative
	$\leftrightarrow$	$\neg(p \vee q) \vee r$	E3(b). Commutative Rule
	$\leftrightarrow$	$((p \vee q) \rightarrow r)$	E1(a). Definition of $\rightarrow$

No questions assigned to the following page.

Problem 2 [10 pts]

Parts (a), (b) and (c) below should all be solved using the formatting used in the study guide. More explicitly
(a) should have the lines numbered, with the premise on the first line and every other line explaining how it was derived from the previous lines. E.g. In the same format as DSP1-4 in the study guide.
(b) should have 8 rows, each one corresponding to one truth setting of p, q, r. E.g., in the same format as Section 6 in the study guide. Hint: Your final table should have 14 columns, including the first 3.
(c) should set specific values of p, q, r and then show, line by line, how the proposition evaluates to 0 for that setting. E.g., in the same format as Section 7 in the study guide.

(a) Given the premise $(p \leftrightarrow q) \wedge (q \leftrightarrow r)$ give a deductive sequence proof that $p \leftrightarrow r$.
Hint: Starting from the premise, prove separately that $p \rightarrow r$ and that $r \rightarrow p$. Combining those two results with the definition of $\leftrightarrow$ proves your conclusion.

(b) Now prove the same thing, i.e., $\Big((p \leftrightarrow q) \wedge (q \leftrightarrow r)\Big) \rightarrow (p \leftrightarrow r)$, *using a truth table.*
Boldface or circle the column corresponding to the rightmost $\rightarrow$, which is the value of the full expression.
We provide a latex template of how to format the truth table with the p, q, r columns filled in. The letters A-K denote the 11 columns that you must fill in.

(c) Prove that $\Big((p \leftrightarrow q) \wedge (q \leftrightarrow r)\Big) \leftrightarrow (p \leftrightarrow r)$ *is NOT a tautology.*

No questions assigned to the following page.

Solution:

(a) [A formatted start of solution is provided.]

1.	$(p \leftrightarrow q) \wedge (q \leftrightarrow r)$	Premise
2.	$p \leftrightarrow q$	I1(a). Left Separation L1
3.	$q \leftrightarrow r$	I1(b). Right Separation L1
4.	$(p \rightarrow q) \wedge (q \rightarrow p)$	E1(b). Definition of $\leftrightarrow$ on L2
5.	$(q \rightarrow r) \wedge (r \rightarrow q)$	E1(b). Definition of $\leftrightarrow$ on L3
6.	$p \rightarrow q$	I1(a). Left Separation L4
7.	$q \rightarrow r$	I1(a). Left Separation L5
8.	$(p \rightarrow q) \wedge (q \rightarrow r)$	Combining L6 & L7
9.	$p \rightarrow r$	I7. Hypothetical Syllogism L8
10.	$q \rightarrow p$	I1(b). Right Separation L4
11.	$r \rightarrow q$	I1(b). Right Separation L5
12.	$(r \rightarrow q) \wedge (q \rightarrow p)$	Combining L11 & L10
13.	$r \rightarrow p$	I7. Hypothetical Syllogism L12
14.	$(p \rightarrow r) \wedge (r \rightarrow p)$	Combining L9 & L13
15.	$p \leftrightarrow r$	E1(b). Definition of $\leftrightarrow$ on L14

(b) [A formatted latex template for the truth table is provided.
Columns A-K need to be filled in.]

p	q	r	$\Big($ (p	$\leftrightarrow$	q)	$\wedge$	(q	$\leftrightarrow$	r) $\Big)$	$\rightarrow$	(p	$\leftrightarrow$	r)
0	0	0	0	1	0	1	0	1	0	**1**	0	1	0
0	0	1	0	1	0	0	0	0	1	**1**	0	0	1
0	1	0	0	0	1	0	1	0	0	**1**	0	1	0
0	1	1	0	0	1	0	1	1	1	**1**	0	0	1
1	0	0	1	0	0	0	0	1	0	**1**	1	0	0
1	0	1	1	0	0	0	0	0	1	**1**	1	1	1
1	1	0	1	1	1	0	1	0	0	**1**	1	0	0
1	1	1	1	1	1	1	1	1	1	**1**	1	1	1

(c) Set $p = r = 1$ and $q = 0$. Then $\Big((p \leftrightarrow q) \wedge (q \leftrightarrow r)\Big) \leftrightarrow (p \leftrightarrow r)$ evaluates to

$$\begin{aligned} \Big((1 \leftrightarrow 0) \wedge (0 \leftrightarrow 1)\Big) \leftrightarrow (1 \leftrightarrow 1) &= (0 \wedge 0) \leftrightarrow 1 && \text{Definition of } \leftrightarrow\text{, applied thrice} \\ &= 0 \leftrightarrow 1 && \text{Definition of } \wedge \\ &= 0 && \text{Definition of } \leftrightarrow \end{aligned}$$

Since the proposition evaluates to 0 for the truth setting $p = r = 1$ and $q = 0$, it is NOT a tautology.

No questions assigned to the following page.

Problem 3. P1.1.8 in book [10 pts]

If S is any finite set of positive naturals, let $sum(S)$ be the natural we get by adding the elements of S together.

(a) There is exactly one set S of positive naturals with $sum(S) = 0$. Are there any other naturals n for which there is exactly one set S with $sum(S) = n$?
If there are not, prove that.
If there are, list them all and argue that you have them all.

(b) List all the sets S with $sum(S) = 10$. (**Hint**: There are exactly ten of them.)

(c) Suppose $S \neq T$ but $sum(S) = sum(T)$. Is it possible that $S \subseteq T$ or that $T \subseteq S$? Why or why not?

Note: The unique set S referenced in (a) with $sum(S) = 0$ is the empty set, i.e., $S = \{\} = \emptyset$. Also, remember that a set either contains an item or doesn't. So, for example, the set $S = \{5, 5, 5, 1\}$ is actually the same as the set $S = \{5, 1\}$ and has $sum(S) = 6$, and not 16.

Solution:

(a) In addition to the empty set (0), there are other naturals n for which there is exactly one set S with sum(S)=n; there is n=1 which is just the set {1}, and n=2 which is just the set {2}. Those however, are all of them; For any value of n greater than 2, there is are at least the sets of the number itself {n}, and the number itself minus 1 and 1 itself {1, }.

(b)

$$\{10\}, \quad \{1,9\}, \quad \{2,8\}, \quad \{3,7\}, \quad \{4,6\},$$
$$\{1,2,7\}, \quad \{1,3,6\}, \quad \{1,4,5\}, \quad \{2,3,5\}, \quad \{1,2,3,4\}$$

(c) It is NOT possible for $S \subseteq T$ or for $T \subseteq S$ if S and T are unique se that have the same sum. This is because subsets require possessing items ONLY from the superset, and since everything in the sets are positive, there's no way for the subset to reach the same sum without additional numbers or by being the entire original set (which is not the case since its outright stated that $S \neq T$).

No questions assigned to the following page.

Problem 4. Modification of Problem 1.1.10 in the book [10 pts]

Let D be a set of dogs, R a set of retrievers, and G a set of golden retrievers. Assume that $G \subseteq R$ and $R \subseteq D$. For each of parts (a)-(d) below first answer (i) and then, starting on a new line, for (ii), provide a complete logical justification as to why your answer is correct. Your justification must use full sentences.

Proving that a number is the minimum number satisfying a property requires proving *both* that the number satisfies the property *and* that no smaller number satisfies that property.

(a) (i) With no other information, what is the minimum number of dogs that must exist? (ii) Justify your answer

(b) Suppose we know that all three sets are different from each other. (i) What is the minimum number of dogs that satisfy this constraint? (ii) Justify your answer.

(c) Suppose we know that no set is empty and that all sets are different from each other. (i) What is the minimum number of dogs that satisfy this constraint? (ii) Justify your answer.

(d) Suppose we know that $D \neq G$ but that the three sets are not all different. (i) What is the minimum number of dogs that satisfy this constraint? (ii) Justify your answer.

Solution:

(a) (i) Zero (0)
(ii) The minimum amount of dogs that must exist is zero (0), because the only information that we are given is the subset order ($G \subseteq R$ and $R \subseteq D$.). With no note of how many dogs are in any given set, we can assume that theoretically, D is just an empty set, R is just the empty subset of that empty set, and G is just the empty subset of that empty subset of that empty set. Since there is no amount of items a set can have that is less than zero, zero MUST be the minimum amount of dogs.

(b) (i) Two (2)
(ii) Given that three sets of dogs are subsets of each other ($G \subseteq R$ and $R \subseteq D$.), and that each set has a different amount of dogs in it, we can safely assume that there are two dogs at the very minimum. This is because the three lowest amount of items a set can have is zero, one, and two; and since the subsets contain only dogs from their supersets, we only need to look at the lowest value the SUPERSET can be, which must be the highest of the three lowest amount of items a set can have (2), since the subsets can't contain items outside of the superset, but the superset CAN contain items outside of the subset.

No questions assigned to the following page.

(c) (i) Three (3)
(ii) Given that three sets of dogs are subsets of each other ($G \subseteq R$ and $R \subseteq D$.), and that each set has a different amount of dogs in it thats more than zero, we can safely assume that there are three dogs at the very minimum. This is because the three lowest amount of items a set can have other than zero is one, two, and three; and since the subsets contain only dogs from their supersets, we only need to look at the lowest value the SUPERSET can be, which must be the highest of the three lowest amount of items a set can have other than zero (3), since the subsets can't contain items outside of the superset, but the superset CAN contain items outside of the subset.

(d) (i) One (1)
(ii) Given that D is not the same set as G, we can safely assume that there is at least 1 dog at the very minimum. This is because assuming the subset is the empty set (the lowest amount of items a set can have), the next lowest amount of items a set can have is 1, and since D is NOT G, it must at LEAST have a count of 1.

No questions assigned to the following page.

Problem 5. P1.4.3 in the book [10 pts]

Another boolean operation is the "NAND" operation. where "x **nand** y" is defined to be $\neg(x \wedge y)$.

(a) Explain why x **nand** x is equivalent to $\neg x$.

(b) Can you find a compound proposition using only the NAND operator that is equivalent to $x \wedge y$? If yes, write it down.

(c) Can you find a compound proposition using only the NAND operator that is equivalent to $x \vee y$? If yes, write it down.

Hint: In parts (b) and (c) you are permitted to use the NAND operator multiple times.
Historical Note: Historically, this had implications in hardware design, since **nand** *gates were easier to physically build than* **and** *or* **or** *gates.*

Solution:

(a) x **nand** x is equivalent to $\neg x$ because the output of both expressions will always be the opposite of x's value (if x is true, the expression is false; if x is false, the expression is true). As a proof it would look like this:

$$\begin{array}{rcll} x \textbf{ nand } x & = & \neg(x \wedge x) & \text{Definition of } \textbf{nand} \\ & = & \neg x & \text{E6(a). Equal Operands} \end{array}$$

(b) The compound proposition $(x$ **nand** $y)$ **nand** $(x$ **nand** $y)$ is equivalent to $x \wedge y$.

$$\begin{array}{rcll} (x \textbf{ nand } y) \textbf{ nand } (x \textbf{ nand } y) & = & \neg\big(\neg(x \wedge y) \wedge \neg(x \wedge y)\big) & \text{Definition of } \textbf{nand} \\ & = & \neg\neg(x \wedge y) & \text{E6(a). Equal Operands} \\ & = & x \wedge y & \text{E9. Double Negative} \end{array}$$

(c) The compound proposition $(x$ **nand** $x)$ **nand** $(y$ **nand** $y)$ is equivalent to $x \vee y$.

$$\begin{array}{rcll} (x \textbf{ nand } x) \textbf{ nand } (y \textbf{ nand } y) & = & \neg(\neg(x \wedge x) \wedge \neg(y \wedge y)) & \text{Definition of } \textbf{nand} \\ & = & \neg(\neg x \wedge \neg y) & \text{E6(a). Equal Operands} \\ & = & x \vee y & \text{E10(b). DeMorgan's Law} \end{array}$$

No questions assigned to the following page.

Problem 6. P1.5.10 in the book [10 pts]

Let $X = \{1, 2, 3, ..., 10\}$ and let I be the set of all **intervals** in X, that is, subsets of the form $\{a, ..., b\}$ for some naturals a and b.

Note: By convention the empty set $\emptyset$ is an interval called the empty interval *(it's what results when $b < a$). Also $a = b$ is allowed. This is called a singleton interval. As examples, $\emptyset, \{3\}, \{3, 4\}, \{5\}, \{3, 4, 5, 6\}$ are all intervals, but $\{3, 5, 6\}$ is NOT an interval.*

For each of (a), (b) and (c) below write the answer (number) in part (i) and in part (ii) explain how you derived your answer.

(a) How many intervals are in the set I? (Don't forget the empty set.)

(b) How many of the intervals are subsets of $\{2, 3, 4, 5, 6\}$?

(c) How many of the intervals are disjoint from $\{2, 3, 4, 5, 6\}$?
Note: An interval Y is disjoint *from $\{2, 3, 4, 5, 6\}$, if $Y \cap \{2, 3, 4, 5, 6\} = \emptyset$.*

Solution:

(a) (i) Fifty Six (56)
(ii) Given the formula $\frac{n(n+1)}{2}$, which calculates the total number of nonempty intervals in a set of n consecutive numbers; plugging in $n = 10$ gives $\frac{10(10+1)}{2} = \frac{10(11)}{2} = \frac{110}{2} = 55$. By convention the additional empty set interval brings the total to 56 $(55 + 1 = 56)$.

(b) (i) Sixteen (16)
(ii) Given the formula $\frac{n(n+1)}{2}$, which calculates the total number of nonempty intervals in a set of n consecutive numbers; plugging in $n = 5$ gives $\frac{5(5+1)}{2} = \frac{5(6)}{2} = \frac{30}{2} = 15$. By convention the additional empty set interval brings the total to 16 $(15+1 = 16)$. Since all intervals are subsets, and every interval within 2,3,4,5,6 is also an interval within X, the total amount of intervals within X that are subsets of 2,3,4,5,6 is just the amount of intervals that 2,3,4,5,6 has (16).

(c) (i) Forty (40)
(ii) The amount of intervals that are disjoint from 2,3,4,5,6 is equal to the total amount of the interval in X MINUS the amount that are subsets of 2,3,4,5,6; in other words, it is $56 - 16 = 40$.

No questions assigned to the following page.

Problem 7. P1.7.6 in the book [10 pts]

Suppose we substituted $a \oplus b$ for p and $a \wedge b$ for q in the contrapositive rule $(p \to q) \leftrightarrow (\neg q \to \neg p)$ to get

$$\Big((a \oplus b) \to (a \wedge b)\Big) \leftrightarrow \Big((\neg a \wedge b) \to (\neg a \oplus b)\Big).$$

(a) Verify that this result is *not* a tautology.
Note: Your verification should be in the same format as those in Section 7 of the Equational and Deductive Sequence Proofs Study Guide.

(b) Why didn't our substitution lead to a valid tautology? Explain.

Solution:

(a) Set $a = 1$ and $b = 0$. Then $\Big((a \oplus b) \to (a \wedge b)\Big) \leftrightarrow \Big((\neg a \wedge b) \to (\neg a \oplus b)\Big)$ evaluates to

$$\begin{array}{lll}
\Big((1 \oplus 0) \to (1 \wedge 0)\Big) \leftrightarrow \Big((\neg 1 \wedge 0) \to (\neg 1 \oplus 0)\Big) & = & \Big((1 \oplus 0) \to (1 \wedge 0)\Big) \leftrightarrow \Big((0 \wedge 0) \to (0 \oplus 0)\Big) \quad \text{Definition of } \neg\text{, applied twice} \\
& = & \Big(1 \to (1 \wedge 0)\Big) \leftrightarrow \Big((0 \wedge 0) \to 0\Big) \quad \text{Definition of } \oplus\text{, applied twice} \\
& = & (1 \to 0) \leftrightarrow (0 \to 0) \quad \text{Definition of } \wedge\text{, applied twice} \\
& = & 0 \leftrightarrow 1 \quad \text{Definition of } \to\text{, applied twice} \\
& = & 0 \quad \text{Definition of } \leftrightarrow
\end{array}$$

Since the proposition evaluates to 0 for the truth setting $a = 1$ and $b = 0$, it is NOT a tautology.

(b) The reason out substitution didn't lead to a valid tautology is that we didn't substitute correctly. The negation operator was supposed to be on the outside of either expression on the right side of the comparison operator, not inside the expression attached to only the variable a. The proper substitution would look like this:

$$\Big((a \oplus b) \to (a \wedge b)\Big) \leftrightarrow \Big(\neg(a \wedge b) \to \neg(a \oplus b)\Big).$$

To prove it, here is the evaluation of the correct substitution:

$$\begin{array}{lll}
\Big((1 \oplus 0) \to (1 \wedge 0)\Big) \leftrightarrow \Big(\neg(1 \wedge 0) \to \neg(1 \oplus 0)\Big) & = & \Big(1 \to (1 \wedge 0)\Big) \leftrightarrow \Big(\neg(1 \wedge 0) \to \neg 1\Big) \quad \text{Definition of } \oplus\text{, applied twice} \\
& = & (1 \to 0) \leftrightarrow (\neg 0 \to \neg 1) \quad \text{Definition of } \wedge\text{, applied twice} \\
& = & (1 \to 0) \leftrightarrow (1 \to 0) \quad \text{Definition of } \neg\text{, applied twice} \\
& = & 0 \leftrightarrow 0 \quad \text{Definition of } \to\text{, applied twice} \\
& = & 1 \quad \text{Definition of } \leftrightarrow
\end{array}$$

No questions assigned to the following page.

Problem 8. P2.1.4 in the book [10 pts]

Let A, B, and C be sets defined in Exercise 2.1.1 in the textbook[1].

(a) Explicitly list the direct products $(A \times B) \times C$ and $A \times (B \times C)$. (Note that an element of $A \times (B \times C)$ is a pair whose first element is in A and whose second element is in $B \times C$.)

(b) If X, Y, and Z are any sets, we generally do not distinguish among $X \times (Y \times Z)$, $(X \times Y) \times Z$, and the set of triples $X \times Y \times Z$. Explain how we may think of an element of each of these sets as an element of one of the others, by describing which elements of each set correspond to which elements of the others.

Solution:

(a)

$$\begin{aligned}(A \times B) \times C = \{&((1,x),\text{cat}),((1,x),\text{dog}),\\ &((1,y),\text{cat}),((1,y),\text{dog}),\\ &((1,z),\text{cat}),((1,z),\text{dog}),\\ &((2,x),\text{cat}),((2,x),\text{dog}),\\ &((2,y),\text{cat}),((2,y),\text{dog}),\\ &((2,z),\text{cat}),((2,z),\text{dog})\}.\end{aligned}$$

$$\begin{aligned}A \times (B \times C) = \{&(1,(x,\text{cat})),(1,(x,\text{dog})),\\ &(1,(y,\text{cat})),(1,(y,\text{dog})),\\ &(1,(z,\text{cat})),(1,(z,\text{dog})),\\ &(2,(x,\text{cat})),(2,(x,\text{dog})),\\ &(2,(y,\text{cat})),(2,(y,\text{dog})),\\ &(2,(z,\text{cat})),(2,(z,\text{dog}))\}.\end{aligned}$$

(b) Although the explicit lists of the direct products come out looking slightly different with the groupings since the parenthesis enclose around different intervals, the core content of each cross product item is the same components in the same order.

$$(x,(y,z)) \leftrightarrow ((x,y),z) \leftrightarrow (x,y,z).$$

[1] i.e., let A be the set $\{1,2\}$, B be the set $\{x,y,z\}$, and C be the set {cat, dog}.

No questions assigned to the following page.

Extra Credit. Modified from Problem 1.6.9 in the book. [10 pts]

In an islander problem[2], we are given a set of statements that are about one another. Each of a set of islanders makes a statement about the truth or falsity of their own or other islanders' statements. Each statement is either true or false, so we have a set of compound propositions with an atomic variable for each.

Often the problem is set up so that *exactly* one setting of the variables matches the variables. That is, each statement whose variable is set to true becomes true, and each one set to false becomes false.

Given the following situations, which islanders, if any, are telling the truth?

For the deductive sequence proofs you may use the statements of Questions 1 (a) and (b) at the start of this homework as accepted rules (that may shorten your proofs).

(a) There are two islanders.
A says "Exactly one of us is lying".
B says "If A is telling the truth, then so am I."

To set this up, the compound proposition for A is $A(a,b) = a \leftrightarrow (a \oplus b)$
and for B is $B(a,b) = b \leftrightarrow (a \rightarrow b)$.

To solve this problem find a truth setting for a, b that makes

$$\Big(A(a,b) \wedge B(a,b)\Big) = \Big(a \leftrightarrow (a \oplus b)\Big) \wedge \Big(b \leftrightarrow (a \rightarrow b)\Big)$$

true. We tell you in advance that there is exactly one setting that makes this proposition true (but you may not use this fact in your derivation).

(i) Draw the complete truth table for this proposition (it has 13 columns in total including the first two) and then, from it, report for each of A and B whether they are telling the truth or lying. The truth table should show that there is exactly one unique setting that staisfies the constraints.

(ii) Now solve the same problem again, but this time using a deductive sequence proof starting with the premises $A(a,b)$ and $B(a,b)$.

(b) There are three islanders.
A says "It is not the case that both the other two are telling the truth."
B says "At least one of the other two is telling the truth."
C says "A is telling the truth, and if B is telling the truth then so am I."

[2]The classic setting for such problems is a remote island where some inhabitants always tell the truth and the others always lie. The logician Raymond Smullyan was famous for designing such puzzles to illustrate logical principles, as in his book *What is the Name of This Book?*.

No questions assigned to the following page.

To set this up, first find the compound propositions for $A(a, b, c)$, $B(a, b, c)$ and $C(a, b, c)$. We start you off by noting that $A(a, b, c) = a \leftrightarrow \neg(b \wedge c)$. In (i) and (ii) write the compound propositions for the other two.

(i) $B(a, b, c) = b \leftrightarrow$ _____

(ii) $C(a, b, c) = c \leftrightarrow$ _____

(iii) Now using a deductive sequence proof with premises $A(a, b, c)$, $B(a, b, c)$ and $C(a, b, c)$ at the top, find a consistent setting for a, b, c that satisfies the three premises. Don't forget, after completing your proof, to show that the setting you found satisfies the constraints.

(iv) For each of A, B and C, say whether they are telling the truth or lying.

No questions assigned to the following page.

Solution:

(a) (i) Draw the truth table and then, from it, state for each of A and B whether they are telling the truth or lying. We start you off by giving the structure of the table and filling in the 8 a and b columns. You only need to fill in the 5 remaining ones. You should conclude that, given the premises there is only one possible setting for a, b. You should then show that that setting satisfies the premises.

a	b	$\Big($ a	$\leftrightarrow$	(	a	$\oplus$	b	) $\Big)$	$\wedge$	$\Big($ b	$\leftrightarrow$	(	a	$\rightarrow$	b	) $\Big)$
0	0	0	1		0	0	0		0	0	0		0	1	0	
0	1	0	0		0	1	1		0	1	1		0	1	1	
1	0	1	1		1	1	0		1	0	1		1	0	0	
1	1	1	0		1	0	1		0	1	1		1	1	1	

An alternative format that you could use would be to fill in 3 truth tables: one for $A(a, b)$, one for $B(a, b)$ and the third for $A(a, b) \wedge B(a, b)$. We also provide starters for that alternative format below.

a	b	a	$\leftrightarrow$	(	a	$\oplus$	b	)
0	0	0	1		0	0	0	
0	1	0	0		0	1	1	
1	0	1	1		1	1	0	
1	1	1	0		1	0	1	

a	b	b	$\leftrightarrow$	(	a	$\rightarrow$	b	)
0	0	0	0		0	1	0	
0	1	1	1		0	1	1	
1	0	0	1		1	0	0	
1	1	1	1		1	1	1	

a	b	$A(a, b)$	$\wedge$	$B(a, b)$
0	0	1	**0**	0
0	1	0	**0**	1
1	0	1	**1**	1
1	1	0	**0**	1

(ii) We start you off with the two premises and the structure of the proof.

(b) (i) $B(a, b, c) = b \leftrightarrow (a \vee c)$

(ii) $C(a, b, c) = c \leftrightarrow \big(a \wedge (b \rightarrow c)\big)$

(iii) We start you off with the top of the proof.

1.	$a \leftrightarrow \neg(b \wedge c)$	Premise
2.	*Fill in* $B(a, b, c)$	Premise
3.	*Fill in* $C(a, b, c)$	Premise
4.	...	Provide Justification
	...	Provide Justification

(iv)